

EDEXCEL INTERNATIONAL A LEVEL

WMA11/01 January 2020 Worked Solutions

Jan 2020 paper rebuilt with the worked solution after each question.

11

paper questions

WMA11

Pure 1

**Single-
paper
solutions**standalone IAL
route**Dr Eslam Ahmed**

Prepared for Dr Eslam Ahmed - eliteigcse.com

P1

PAST PAPER

WMA11/01 January 2020

Jan 2020 | 11 questions | 75 marks

11

questions

75

marks

Answers

worked solution
after each
question

Question 1**Integration**

1. Find, in simplest form,

$$\int \left(\frac{8x^3}{3} - \frac{1}{2\sqrt{x}} - 5 \right) dx$$

(4)

(Total 4 marks)

Worked Solution - Question 1

Primary topic

1. Rewrite the integrand using powers

$$\frac{8x^3}{3} - \frac{1}{2\sqrt{x}} - 5 = \frac{8}{3}x^3 - \frac{1}{2}x^{-1/2} - 5.$$

2. Integrate term by term

$$\int \frac{8}{3}x^3 dx = \frac{2}{3}x^4, \int -\frac{1}{2}x^{-1/2} dx = -x^{1/2}, \text{ and } \int -5 dx = -5x.$$

3. Include the constant

$$\text{Therefore } \int \left(\frac{8x^3}{3} - \frac{1}{2\sqrt{x}} - 5 \right) dx = \frac{2}{3}x^4 - \sqrt{x} - 5x + c.$$

Final answer

$$\frac{2}{3}x^4 - \sqrt{x} - 5x + c.$$

Question 2

Indices & Surds

2. Given $y = 3^x$, express each of the following in terms of y . Write each expression in its simplest form.

(a) 3^{3x} (1)

(b) $\frac{1}{3^{x-2}}$ (2)

(c) $\frac{81}{9^{2-3x}}$ (2)

(Total 5 marks)

Worked Solution - Question 2

Primary topic

1. Use $y = 3^x$

Since $y = 3^x$, powers of 3^x can be replaced by powers of y .

2. Part (a)

$$3^{3x} = (3^x)^3 = y^3.$$

3. Part (b)

$$\frac{1}{3^{x-2}} = 3^{2-x} = \frac{3^2}{3^x} = \frac{9}{y}.$$

4. Part (c)

$$\frac{81}{9^{2-3x}} = \frac{3^4}{(3^2)^{2-3x}} = \frac{3^4}{3^{4-6x}} = 3^{6x} = (3^x)^6 = y^6.$$

Final answer

(a) y^3 .

(b) $\frac{9}{y}$.

(c) y^6 .

Question 3

Differentiation

3.

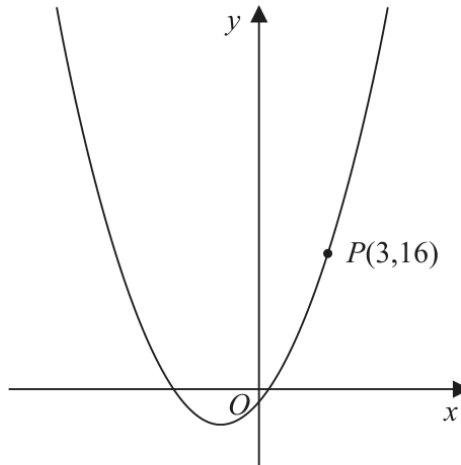


Figure 1

Figure 1 shows part of the curve with equation $y = x^2 + 3x - 2$

The point $P(3, 16)$ lies on the curve.

(a) Find the gradient of the tangent to the curve at P .

(2)

The point Q with x coordinate $3 + h$ also lies on the curve.

(b) Find, in terms of h , the gradient of the line PQ . Write your answer in simplest form.

(3)

(c) Explain briefly the relationship between the answer to (b) and the answer to (a).

(1)

(Total 6 marks)

Worked Solution - Question 3

1. Differentiate for the tangent gradient

For $y = x^2 + 3x - 2$, $\frac{dy}{dx} = 2x + 3$. At $x = 3$, the gradient is $2(3) + 3 = 9$.

2. Find the y-coordinate of Q

For $x_Q = 3 + h$, $y_Q = (3 + h)^2 + 3(3 + h) - 2 = 16 + 9h + h^2$.

3. Calculate the gradient of PQ

$$\text{gradient } PQ = \frac{(16 + 9h + h^2) - 16}{(3 + h) - 3} = \frac{9h + h^2}{h} = 9 + h.$$

4. Explain the relationship

As $h \rightarrow 0$, point Q approaches P , and the chord gradient $9 + h$ approaches the tangent gradient 9.

Final answer

(a) 9.

(b) $9 + h$.

(c) As $h \rightarrow 0$, $9 + h \rightarrow 9$.

Question 4

Radians

4.

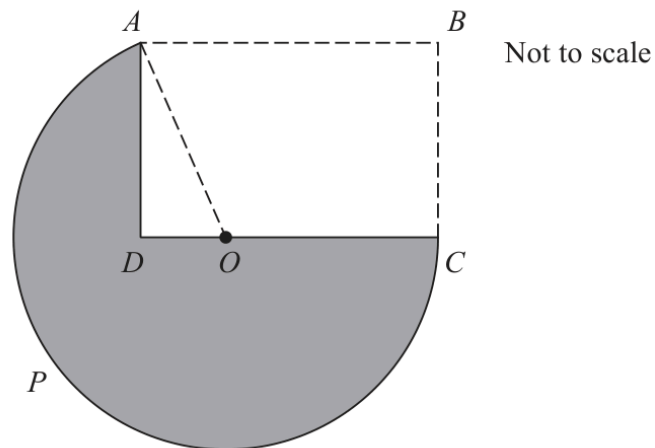


Figure 2

Figure 2 shows the plan view of a house $ABCD$ and a lawn $APCDA$.

$ABCD$ is a rectangle with $AB = 16$ m.

$APCOA$ is a sector of a circle centre O with radius 12 m.

The point O lies on the line DC , as shown in Figure 2.

(a) Show that the size of angle AOD is 1.231 radians to 3 decimal places.

(2)

The lawn $APCDA$ is shown shaded in Figure 2.

(b) Find the area of the lawn, in m^2 , to one decimal place.

(4)

(c) Find the perimeter of the lawn, in metres, to one decimal place.

(3)

(Total 9 marks)

Worked Solution - Question 4

Primary topic

1. Find angle AOD

Since $OC = 12$ and $DC = AB = 16$, $OD = 4$. In right triangle AOD ,
 $\cos \angle AOD = \frac{OD}{OA} = \frac{4}{12} = \frac{1}{3}$. Hence $\angle AOD = \cos^{-1} \left(\frac{1}{3} \right) = 1.231$ radians
to 3 d.p.

2. Identify the major sector used for the lawn

The shaded lawn uses the major sector from A to C , so its angle is $\pi + 1.231$ radians.

3. Subtract the unshaded triangle

$AD = \sqrt{12^2 - 4^2} = 8\sqrt{2}$. Area of major sector = $\frac{1}{2}(12)^2(\pi + 1.231)$. Area of triangle $AOD = \frac{1}{2}(4)(8\sqrt{2})$.

4. Find the lawn area

Area = $\frac{1}{2}(12)^2(\pi + 1.231) - \frac{1}{2}(4)(8\sqrt{2}) = 292.2 \text{ m}^2$ to 1 d.p.

5. Find the lawn perimeter

Perimeter = major arc + $CD + AD = 12(\pi + 1.231) + 16 + 8\sqrt{2} = 79.8 \text{ m}$
to 1 d.p.

Final answer

(a) $\angle AOD = 1.231 \text{ rad.}$

(b) 292.2 m^2 .

(c) 79.8 m.

Question 5**Quadratics**

5. (a) Find, using algebra, all solutions of

$$20x^3 - 50x^2 - 30x = 0 \quad (3)$$

- (b) Hence find all real solutions of

$$20(y+3)^{\frac{3}{2}} - 50(y+3) - 30(y+3)^{\frac{1}{2}} = 0 \quad (4)$$

(Total 7 marks)

Worked Solution - Question 5

Primary topic

1. Factorise the cubic

$$20x^3 - 50x^2 - 30x = 10x(2x^2 - 5x - 3) = 10x(2x + 1)(x - 3).$$

2. Solve part (a)

$$10x(2x + 1)(x - 3) = 0, \text{ so } x = 0, x = -\frac{1}{2}, \text{ or } x = 3.$$

3. Use the same structure for part (b)

Let $u = (y + 3)^{1/2}$. Then the equation becomes $20u^3 - 50u^2 - 30u = 0$, so $u = 0$, $u = -\frac{1}{2}$, or $u = 3$.

4. Reject the invalid square-root value

Since $u = \sqrt{y + 3} \geq 0$, reject $u = -\frac{1}{2}$. If $u = 0$, then $y = -3$. If $u = 3$, then $y + 3 = 9$, so $y = 6$.

Final answer

$$(a) x = 0, -\frac{1}{2}, 3.$$

$$(b) y = -3, 6.$$

Question 6

Straight Line

6. The line l_1 has equation $3x - 4y + 20 = 0$

The line l_2 cuts the x -axis at $R(8,0)$ and is parallel to l_1

- (a) Find the equation of l_2 , writing your answer in the form $ax + by + c = 0$, where a , b and c are integers to be found.

(3)

The line l_1 cuts the x -axis at P and the y -axis at Q .

Given that $PQRS$ is a parallelogram, find

- (b) the area of $PQRS$,

(3)

- (c) the coordinates of S .

(2)

(Total 8 marks)

Worked Solution - Question 6

1. Find the gradient of l_1

$3x - 4y + 20 = 0$ gives $y = \frac{3}{4}x + 5$, so the gradient is $\frac{3}{4}$.

2. Find l_2 through R

Line l_2 is parallel to l_1 and passes through $R(8, 0)$, so $y = \frac{3}{4}(x - 8) = \frac{3}{4}x - 6$.
Therefore $3x - 4y - 24 = 0$.

3. Find P and Q

For l_1 , the x-intercept is $P(-\frac{20}{3}, 0)$ and the y-intercept is $Q = (0, 5)$.

4. Find the area of PQRS

Base $PR = 8 - (-\frac{20}{3}) = \frac{44}{3}$ and height $OQ = 5$. Area = $\frac{44}{3} \cdot 5 = \frac{220}{3}$.

5. Find S using a vector

In parallelogram $PQRS$, $\vec{PS} = \vec{QR} = (8, -5)$. Hence
 $S = P + (8, -5) = (-\frac{20}{3} + 8, -5) = (\frac{4}{3}, -5)$.

Final answer

(a) $3x - 4y - 24 = 0$.

(b) $\frac{220}{3}$.

(c) $S = (\frac{4}{3}, -5)$.

Question 7

Trigonometric Functions

7.

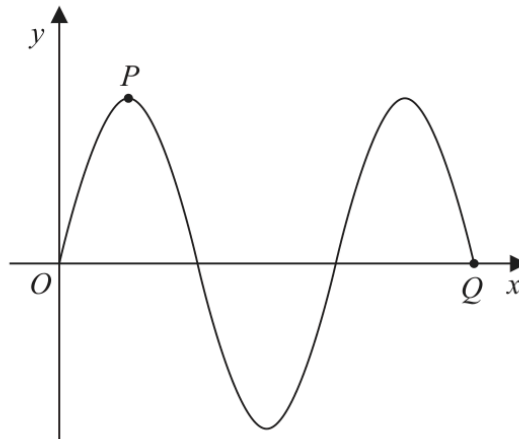


Figure 3

Figure 3 shows part of the curve C_1 with equation $y = 3 \sin x$, where x is measured in degrees.

The point P and the point Q lie on C_1 and are shown in Figure 3.

(a) State

- (i) the coordinates of P ,
- (ii) the coordinates of Q .

(3)

A different curve C_2 has equation $y = 3 \sin x + k$, where k is a constant.

The curve C_2 has a maximum y value of 10

The point R is the minimum point on C_2 with the smallest positive x coordinate.

(b) State the coordinates of R .

(2)

(Total 5 marks)

Worked Solution - Question 7

Primary topic

1. Read P from $y = 3 \sin x$

The first maximum of $y = 3 \sin x$ occurs at $x = 90^\circ$, with $y = 3$. So $P = (90^\circ, 3)$.

2. Read Q from the next shown zero

After the second maximum, the curve returns to the x-axis at $x = 540^\circ$. So $Q = (540^\circ, 0)$.

3. Find the vertical shift

For C_2 , $y = 3 \sin x + k$. Its maximum is $3 + k = 10$, so $k = 7$.

4. Find the minimum point R

The minimum value is $-3 + 7 = 4$. The smallest positive x-coordinate for a sine minimum is 270° , so $R = (270^\circ, 4)$.

Final answer

(a)(i) $P = (90^\circ, 3)$. (a)(ii) $Q = (540^\circ, 0)$.

(b) $R = (270^\circ, 4)$.

Question 8

Inequalities

-
8. The straight line l has equation $y = k(2x - 1)$, where k is a constant.

The curve C has equation $y = x^2 + 2x + 11$

Find the set of values of k for which l does not cross or touch C .

(6)

(Total 6 marks)

Worked Solution - Question 8

1. Equate the line and curve

$$k(2x - 1) = x^2 + 2x + 11, \text{ so } x^2 + (2 - 2k)x + (11 + k) = 0.$$

2. Use the no-intersection condition

For the line not to cross or touch the curve, the quadratic in x must have no real roots, so its discriminant is less than zero.

3. Form the inequality in k

$$(2 - 2k)^2 - 4(1)(11 + k) < 0. \text{ Dividing by 4 gives } (1 - k)^2 - (11 + k) < 0, \text{ so } k^2 - 3k - 10 < 0.$$

4. Solve the quadratic inequality

$$(k - 5)(k + 2) < 0, \text{ so } k \text{ lies between the two roots: } -2$$

Final answer -2

Question 9

Differentiation

9. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

A curve has equation

$$y = \frac{4x^2 + 9}{2\sqrt{x}} \quad x > 0$$

Find the x coordinate of the point on the curve at which $\frac{dy}{dx} = 0$

(6)

(Total 6 marks)

Worked Solution - Question 9

1. Rewrite the function

$$y = \frac{4x^2 + 9}{2\sqrt{x}} = 2x^{3/2} + \frac{9}{2}x^{-1/2}.$$

2. Differentiate

$$\frac{dy}{dx} = 3x^{1/2} - \frac{9}{4}x^{-3/2}.$$

3. Set the derivative equal to zero

$$3x^{1/2} - \frac{9}{4}x^{-3/2} = 0, \text{ so } 3\sqrt{x} = \frac{9}{4x^{3/2}}.$$

4. Solve for x

Multiply by $4x^{3/2}$: $12x^2 = 9$. Hence $x^2 = \frac{3}{4}$. Since $x > 0$, $x = \frac{\sqrt{3}}{2}$.

Final answer

$$x = \frac{\sqrt{3}}{2}.$$

Question 10

Graphs of Functions

10. The curve C_1 has equation $y = f(x)$, where

$$f(x) = (4x - 3)(x - 5)^2$$

- (a) Sketch C_1 showing the coordinates of any point where the curve touches or crosses the coordinate axes.

(3)

- (b) Hence or otherwise

(i) find the values of x for which $f\left(\frac{1}{4}x\right) = 0$

- (ii) find the value of the constant p such that the curve with equation $y = f(x) + p$ passes through the origin.

(2)

A second curve C_2 has equation $y = g(x)$, where $g(x) = f(x + 1)$

- (c) (i) Find, in simplest form, $g(x)$. You may leave your answer in a factorised form.

- (ii) Hence, or otherwise, find the y intercept of curve C_2

(3)

(Total 8 marks)

Worked Solution - Question 10

Primary topic

1. Use the factorised form for the sketch

$f(x) = (4x - 3)(x - 5)^2$. The curve crosses the x-axis at $x = \frac{3}{4}$ and touches the x-axis at $x = 5$. The y-intercept is $f(0) = (-3)(25) = -75$.

2. Solve $f(x/4)=0$

For $f\left(\frac{x}{4}\right) = 0$, either $4\left(\frac{x}{4}\right) - 3 = 0$ or $\frac{x}{4} - 5 = 0$. Therefore $x = 3$ or $x = 20$.

3. Find p

The curve $y = f(x) + p$ passes through the origin, so $0 = f(0) + p = -75 + p$. Hence $p = 75$.

4. Find g(x)

$$g(x) = f(x + 1) = (4(x + 1) - 3)((x + 1) - 5)^2 = (4x + 1)(x - 4)^2.$$

5. Find the y-intercept of C2

At $x = 0$, $g(0) = (1)(16) = 16$. So the y-intercept is 16.

Final answer

Sketch: crosses at $\left(\frac{3}{4}, 0\right)$, touches at $(5, 0)$, crosses y-axis at $(0, -75)$.

Also $x = 3, 20$, $p = 75$, $g(x) = (4x + 1)(x - 4)^2$, y-intercept 16.

Question 11

Integration

11. A curve has equation $y = f(x)$, where

$$f''(x) = \frac{6}{\sqrt{x^3}} + x \quad x > 0$$

The point $P(4, -50)$ lies on the curve.

Given that $f'(x) = -4$ at P ,

- (a) find the equation of the normal at P , writing your answer in the form $y = mx + c$, where m and c are constants,

(3)

- (b) find $f(x)$.

(8)

(Total 11 marks)

Worked Solution - Question 11

1. Find the normal gradient

At P , $f'(4) = -4$, so the tangent gradient is -4 . The normal gradient is therefore $\frac{1}{4}$.

2. Find the normal equation

Using $P(4, -50)$: $y + 50 = \frac{1}{4}(x - 4)$, so $y = \frac{1}{4}x - 51$.

3. Integrate f'' to find f'

$f''(x) = 6x^{-3/2} + x$. Therefore $f'(x) = -12x^{-1/2} + \frac{1}{2}x^2 + k$.

4. Use $f'(4) = -4$

$-4 = -\frac{12}{2} + \frac{1}{2}(16) + k = -6 + 8 + k = 2 + k$, so $k = -6$.

5. Integrate again

$f(x) = -24x^{1/2} + \frac{x^3}{6} - 6x + d$.

6. Use the point on the curve

Substitute $P(4, -50)$:

$-50 = -24(2) + \frac{64}{6} - 24 + d = -72 + \frac{32}{3} + d = -\frac{184}{3} + d$. Hence $d = \frac{34}{3}$.

Final answer

$$(a) y = \frac{1}{4}x - 51.$$

$$(b) f(x) = -24\sqrt{x} + \frac{x^3}{6} - 6x + \frac{34}{3}.$$